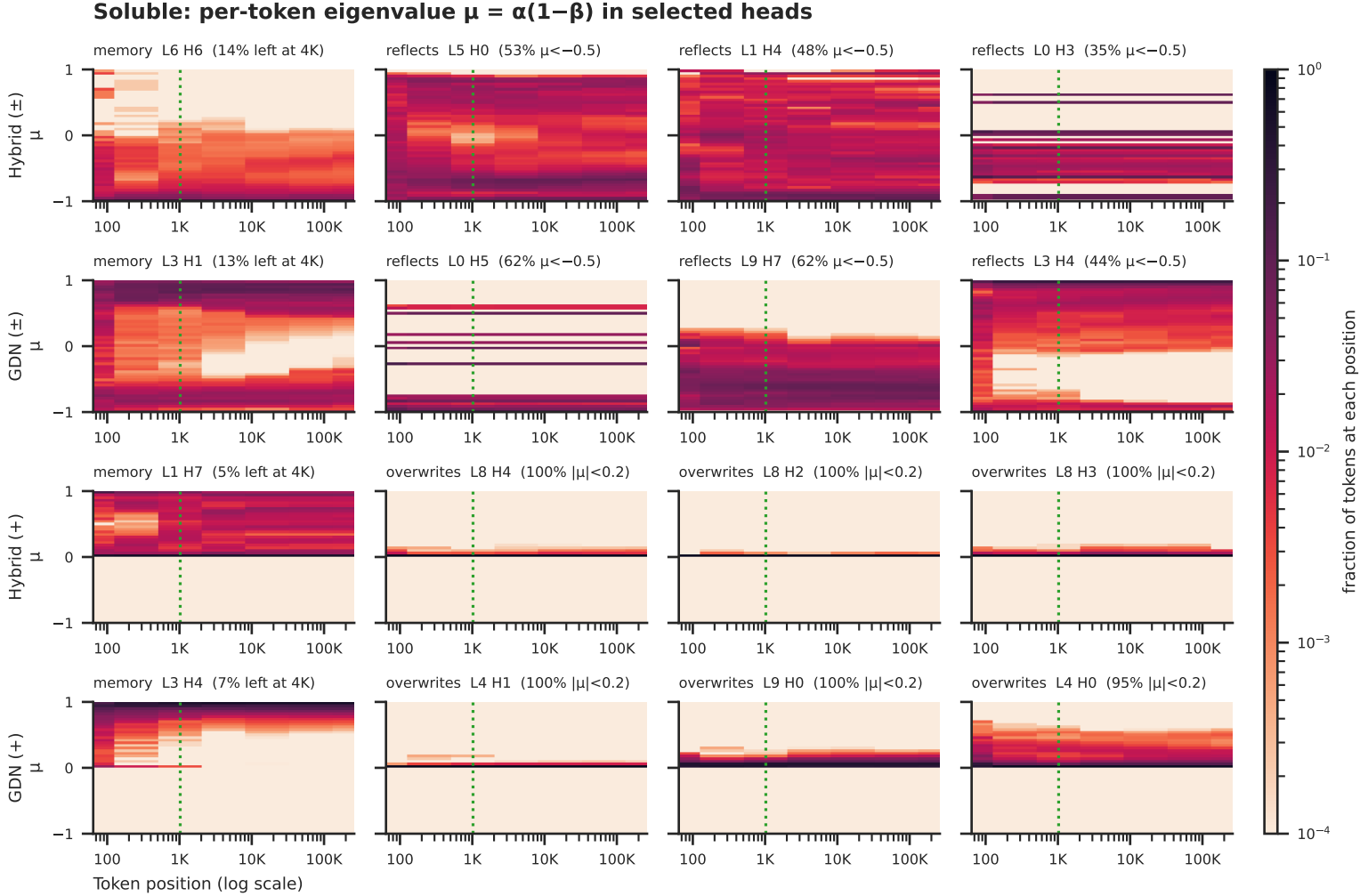

Soluble: per-token eigenvalue μ = α(1−β) in selected heads
memory L6 H6 (14% left at 4K)
reflects L5 H0 (53% μ<−0.5)
reflects L1 H4 (48% μ<−0.5)
reflects L0 H3 (35% μ<−0.5)
memory L3 H1 (13% left at 4K)
reflects L0 H5 (62% μ<−0.5)
reflects L9 H7 (62% μ<−0.5)
reflects L3 H4 (44% μ<−0.5)
memory L1 H7 (5% left at 4K)
overwrites L8 H4 (100% |μ|<0.2)
overwrites L8 H2 (100% |μ|<0.2)
overwrites L8 H3 (100% |μ|<0.2)
memory L3 H4 (7% left at 4K)
overwrites L4 H1 (100% |μ|<0.2)
overwrites L9 H0 (100% |μ|<0.2)
overwrites L4 H0 (95% |μ|<0.2)
Hybrid (±)
GDN (±)
Hybrid (+)
GDN (+)
μ
1
0
−1
100
1K
10K
100K
Token position (log scale)
fraction of tokens at each position
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}